CLASS 10 · CBSE · NCERT MATHEMATICS · NCERT SOLUTIONS

Chapter 14: Probability

Step-by-step solutions to the textbook exercises: 25 questions

Questions are numbered as in the textbook. Try each one yourself before reading the solution.

1 Exercise 14.1

Q1 Fill in the blanks: (i) $P(E) + P(\text{not } E) =$ ___. (ii) An event that can never happen has probability ___ and is called an ___ event. (iii) An event that is bound to happen has probability ___ and is called a ___ event. (iv) Adding the probabilities of all the elementary events of an experiment always gives ___. (v) The probability of any event lies between ___ and ___, both included.

SOLUTION

Answer: (i) 1 (ii) 0, an impossible event (iii) 1, a sure (certain) event (iv) 1 (v) 0 and 1

Q2 Which experiments have equally likely outcomes? Explain. (i) A driver tries to start a car; it starts or does not start. (ii) A player shoots a basketball; the shot is made or missed. (iii) A true-false question is answered; the answer is right or wrong. (iv) A baby is born; it is a boy or a girl.

SOLUTION

Outcomes are equally likely when there is no reason to expect one more than another.

(i) Not equally likely. Whether the car starts depends on its condition, so the two outcomes do not have the same chance.

(ii) Not equally likely. The result depends on the skill of the player.

(iii) Equally likely. A guess at a true-false question is right or wrong with the same chance.

(iv) Equally likely. A baby is a boy or a girl with the same chance.

Answer: (iii) and (iv) have equally likely outcomes

Q3 Why is tossing a coin a fair way to decide which team gets the ball at the start of a football game?

SOLUTION

A coin has only two outcomes, head and tail, and they are equally likely. Neither team can predict or control the result.

Answer: because each team has the same probability, $\frac{1}{2}$, of winning the toss

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Q4 Which of these values is impossible as the probability of an event? (A) $\frac{2}{3}$ (B) -1.5 (C) 15% (D) 0.7

SOLUTION

A probability always lies from 0 to 1. $\frac{2}{3}$, $15\% = 0.15$ and 0.7 are in this range, but -1.5 is negative.

Answer: (B) -1.5

Q5 If $P(E) = 0.05$, find the probability of "not E".

SOLUTION

$P(\text{not } E) = 1 - P(E) = 1 - 0.05.$

Answer: 0.95

Q6 A bag contains lemon flavoured candies only. One candy is taken out without looking. Find the probability that it is (i) an orange flavoured candy (ii) a lemon flavoured candy.

SOLUTION

(i) There is no orange flavoured candy in the bag, so this cannot happen: the probability is 0.

(ii) Every candy is lemon flavoured, so this is certain: the probability is 1.

Answer: (i) 0 (ii) 1

Q7 In a group of 3 students, the probability that 2 students do not have the same birthday is 0.992. Find the probability that the 2 students have the same birthday.

SOLUTION

The two events are complements: $P(\text{same birthday}) = 1 - 0.992.$

Answer: 0.008

Q8 A bag has 3 red balls and 5 black balls. One ball is drawn at random. Find the probability that it is (i) red (ii) not red.

SOLUTION

Total balls $= 8$. (i) $P(\text{red}) = \frac{3}{8}$. (ii) $P(\text{not red}) = 1 - \frac{3}{8} = \frac{5}{8}$.

Answer: (i) $\frac{3}{8}$ (ii) $\frac{5}{8}$

Q9 A box has 5 red, 8 white and 4 green marbles. One marble is taken out at random. Find the probability that it is (i) red (ii) white (iii) not green.

SOLUTION

Total marbles $= 17$. (i) $\frac{5}{17}$ (ii) $\frac{8}{17}$ (iii) $\frac{5+8}{17} = \frac{13}{17}$.

Answer: (i) $\frac{5}{17}$ (ii) $\frac{8}{17}$ (iii) $\frac{13}{17}$

Q10 A piggy bank has a hundred 50p coins, fifty ₹1 coins, twenty ₹2 coins and ten ₹5 coins. One coin falls out, each being equally likely. Find the probability that it (i) is a 50p coin (ii) is not a ₹5 coin.

SOLUTION

Total coins $= 100 + 50 + 20 + 10 = 180$.
(i) $\frac{100}{180} = \frac{5}{9}$. (ii) $\frac{180 - 10}{180} = \frac{17}{18}$.
Answer: (i) $\frac{5}{9}$ (ii) $\frac{17}{18}$

Q11 A fish is taken at random from a tank with 5 male fish and 8 female fish. Find the probability that it is a male fish.

SOLUTION

Total fish $= 13$, of which 5 are male.
Answer: $\frac{5}{13}$

Q12 An arrow is spun and comes to rest pointing at one of the numbers 1 to 8, all equally likely. Find the probability that it points at (i) 8 (ii) an odd number (iii) a number greater than 2 (iv) a number less than 9.

SOLUTION

There are 8 outcomes. (i) One outcome: $\frac{1}{8}$. (ii) 1, 3, 5, 7: $\frac{4}{8} = \frac{1}{2}$. (iii) 3, 4, 5, 6, 7, 8: $\frac{6}{8} = \frac{3}{4}$. (iv) All 8 numbers: $\frac{8}{8} = 1$.
Answer: (i) $\frac{1}{8}$ (ii) $\frac{1}{2}$ (iii) $\frac{3}{4}$ (iv) 1

Q13 A die is rolled once. What is the probability that the number shown is (i) prime (ii) strictly between 2 and 6 (iii) odd?

SOLUTION

There are 6 outcomes. (i) 2, 3, 5: $\frac{3}{6} = \frac{1}{2}$. (ii) 3, 4, 5: $\frac{3}{6} = \frac{1}{2}$. (iii) 1, 3, 5: $\frac{3}{6} = \frac{1}{2}$.
Answer: $\frac{1}{2}$ in each case

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Q14 A card is picked at random from a well-shuffled pack of 52. Find the probability that it is (i) a red king (ii) a face card (iii) a red face card (iv) the jack of hearts (v) a spade (vi) the queen of diamonds.

SOLUTION

(i) There are 2 red kings: $\frac{2}{52} = \frac{1}{26}$.

(ii) There are 12 face cards (jack, queen, king in 4 suits): $\frac{12}{52} = \frac{3}{13}$.

(iii) There are 6 red face cards: $\frac{6}{52} = \frac{3}{26}$.

(iv) There is 1 jack of hearts: $\frac{1}{52}$.

(v) There are 13 spades: $\frac{13}{52} = \frac{1}{4}$.

(vi) There is 1 queen of diamonds: $\frac{1}{52}$.

Answer: (i) $\frac{1}{26}$ (ii) $\frac{3}{13}$ (iii) $\frac{3}{26}$ (iv) $\frac{1}{52}$ (v) $\frac{1}{4}$ (vi) $\frac{1}{52}$

Q15 Five cards, the ten, jack, queen, king and ace of diamonds, are shuffled face down and one is picked at random. (i) Find the probability that it is the queen. (ii) If the queen is drawn and put aside, find the probability that the second card picked is (a) an ace (b) a queen.

SOLUTION

(i) One queen among 5 cards: $\frac{1}{5}$.

(ii) Four cards remain: ten, jack, king, ace. (a) One ace: $\frac{1}{4}$. (b) No queen is left: 0.

Answer: (i) $\frac{1}{5}$ (ii) (a) $\frac{1}{4}$ (b) 0

Q16 12 defective pens are mixed with 132 good ones. One pen is taken at random. Find the probability that it is a good one.

SOLUTION

Total pens $= 144$. $P(\text{good}) = \frac{132}{144} = \frac{11}{12}$.

Answer: $\frac{11}{12}$

Q17 (i) A lot of 20 bulbs has 4 defective ones. One bulb is drawn at random. Find the probability that it is defective. (ii) The bulb drawn in (i) is not defective and is not replaced. One more bulb is drawn from the rest. Find the probability that it is not defective.

SOLUTION

(i) $\frac{4}{20} = \frac{1}{5}$.

(ii) 19 bulbs remain, of which $16 - 1 = 15$ are not defective: $\frac{15}{19}$.

Answer: (i) $\frac{1}{5}$ (ii) $\frac{15}{19}$

Q18 A box has 90 discs numbered 1 to 90. One disc is drawn at random. Find the probability that it bears (i) a two-digit number (ii) a perfect square (iii) a number divisible by 5.

SOLUTION

(i) Two-digit numbers are 10 to 90, which is 81 numbers: $\frac{81}{90} = \frac{9}{10}$.

(ii) Perfect squares are $1, 4, 9, \ldots, 81$, which is 9 numbers: $\frac{9}{90} = \frac{1}{10}$.

(iii) Multiples of 5 are $5, 10, \ldots, 90$, which is 18 numbers: $\frac{18}{90} = \frac{1}{5}$.

Answer: (i) $\frac{9}{10}$ (ii) $\frac{1}{10}$ (iii) $\frac{1}{5}$

Q19 The six faces of a die show the letters A, B, C, D, E, A. The die is thrown once. Find the probability of getting (i) A (ii) D.

SOLUTION

(i) Two faces show A: $\frac{2}{6} = \frac{1}{3}$. (ii) One face shows D: $\frac{1}{6}$.

Answer: (i) $\frac{1}{3}$ (ii) $\frac{1}{6}$

Q20* A die is dropped at random on a rectangle 3 m long and 2 m wide. Find the probability that it lands inside a circle of diameter 1 m drawn in the rectangle.

SOLUTION

Area of the rectangle $= 3 \times 2 = 6$ m^2. Area of the circle $= \pi \left(\frac{1}{2}\right)^2 = \frac{\pi}{4}$ m^2.

Probability $= \frac{\pi/4}{6} = \frac{\pi}{24}$.

Answer: $\frac{\pi}{24}$

Q21 A lot has 144 ball pens, of which 20 are defective. A buyer takes a pen only if it is good. The shopkeeper draws one pen at random. Find the probability that the buyer (i) will buy it (ii) will not buy it.

SOLUTION

Good pens $= 144 - 20 = 124$. (i) $\frac{124}{144} = \frac{31}{36}$. (ii) $\frac{20}{144} = \frac{5}{36}$.

Answer: (i) $\frac{31}{36}$ (ii) $\frac{5}{36}$

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Q22 Two dice are thrown. (i) Complete the table of the probability of each sum from 2 to 12 (the textbook gives $\frac{1}{36}$ for 2, $\frac{5}{36}$ for 8 and $\frac{1}{36}$ for 12). (ii) A student says the 11 sums 2, 3, ..., 12 each have probability $\frac{1}{11}$. Is this right?

SOLUTION

(i) There are 36 equally likely outcomes. Count the pairs that give each sum: for example, sum 4 comes from $(1,3)$, $(2,2)$, $(3,1)$.

Sum	2	3	4	5	6	7	8	9	10	11	12
Number of ways	1	2	3	4	5	6	5	4	3	2	1
Probability	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

(ii) No. The 11 sums are not equally likely: a sum of 7 can come in 6 ways but a sum of 2 in only 1 way. So their probabilities are different, as the table shows.

Answer: (i) as in the table (ii) the argument is not correct

Q23 A coin is tossed 3 times. A player wins if all three tosses give the same result and loses otherwise. Find the probability that the player loses.

SOLUTION

The 8 equally likely outcomes are HHH, HHT, HTH, THH, HTT, THT, TTH, TTT.

The player wins with HHH or TTT, so the player loses in the other 6 outcomes: $\frac{6}{8} = \frac{3}{4}$.

Answer: $\frac{3}{4}$

Q24 A die is thrown twice. Find the probability that (i) 5 does not come up either time (ii) 5 comes up at least once.

SOLUTION

There are $6 \times 6 = 36$ outcomes.

(i) Each throw can show any of the 5 numbers other than 5, so there are $5 \times 5 = 25$ such outcomes: $\frac{25}{36}$.

(ii) This is the complement of (i): $1 - \frac{25}{36} = \frac{11}{36}$.

Answer: (i) $\frac{25}{36}$ **(ii)** $\frac{11}{36}$

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Q25 Which arguments are correct? Give reasons. (i) When two coins are tossed there are three outcomes, two heads, two tails or one of each, so each has probability $\frac{1}{3}$. (ii) When a die is thrown there are two outcomes, an odd number or an even number, so the probability of an odd number is $\frac{1}{2}$.

SOLUTION

(i) Not correct. The equally likely outcomes are HH, HT, TH and TT. "One of each" covers both HT and TH, so its probability is $\frac{2}{4} = \frac{1}{2}$, while two heads and two tails each have probability $\frac{1}{4}$.

(ii) Correct. There are 3 odd numbers (1, 3, 5) and 3 even numbers (2, 4, 6) among 6 equally likely outcomes, so "odd" and "even" are equally likely and $P(\text{odd}) = \frac{3}{6} = \frac{1}{2}$.

Answer: (i) is not correct and (ii) is correct

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